Experiment No: 03

Name of The Experiment: Verification of Ohms Law.

Objectives:

a) To verify the ohm's law.

b) To determine the unknown resistance of the given material of the wire.

Theory:

The fundamental relationship among voltage, current, and resistance was discovered by Georg Simon Ohm, which is known as Ohm's law. Ohm's law states that,

"If temperature and other physical conditions of a conductor remain unchanged then current flowing through the conductor is directly proportional to the voltage difference between two terminals of the conductors".

Mathematically,

$I \alpha V$

$I=GV$ [G is a constant called conductance]

$G=1/R$ [R is called resistance]

So that,

$I=V/R$

$R=V/I$

Let I=y, V=x and 1/R=m then equation (1) can be written as y=mx, which is the equation of straight line passing through the origin. Therefore, if we apply different voltages across a resistor and measure resulting voltage and current using voltmeter and ammeter respectively, and put them in a graph the result will be a straight line. This verifies the ohm's law. Again slope of this straight line is calculated as V/I. This slope is equal to1/R. Where, R is the value of resistance of the resistor. Therefore, calculating the slope from the graph we can calculate the value of unknown resistance using above relation.

Theoretical Diagram:

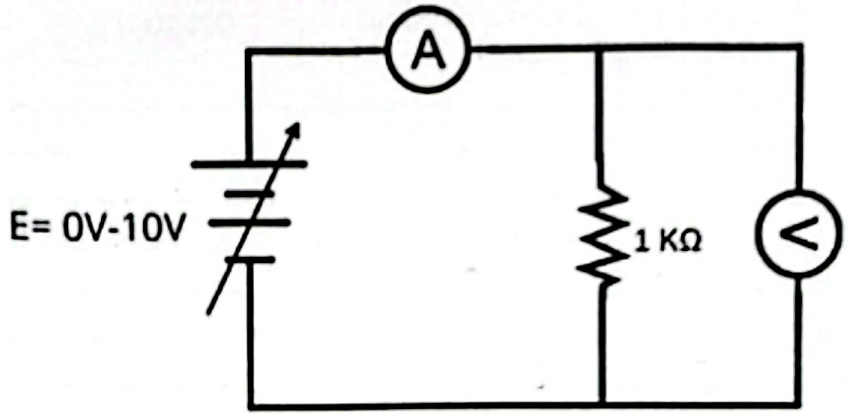


Fig 3(a): Ohm's Law verification circuit

Apparatus:

i. Variable DC power supply
ii. Breadboard
iii. Resistors
iv. Voltmeter
v. Ammeter
vi. Connecting wires

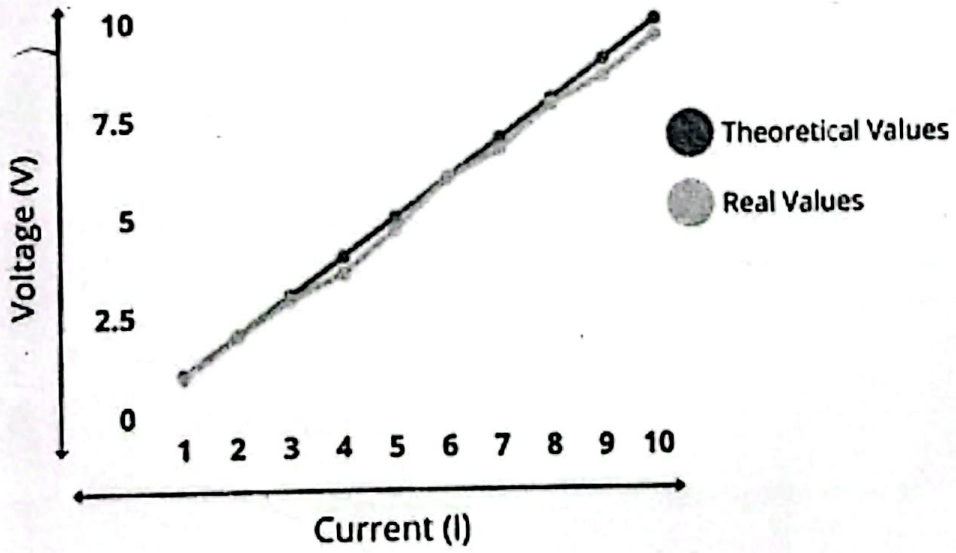
Procedure:

a) Connect all components on the bread board according to the circuit of Fig 3(a) on the bread board.
b) Turn on the dc power supply.
c) Set the dc power supply voltage to zero volt.
d) Measure the voltage and current using voltmeter and ammeter respectively.
e) Increase the voltage of power supply by 1 volt and repeat the above step for few times.

Experimental Table:

OBS	Ammeter reading, I(ampere)	Voltmeter reading V(volt)	Resistance, R=V/I (ohm)
1.			
2.			
3.			
4.			
5.			
6.			
7.			
8.			
9.			
10.			
11.			

Graph:



Results:

Ohm's law is verified and the calculated value of unknown resistance is……

Discussion:

The V-I graph is not exactly straight line. It is deviated slightly from the ideal graph. Moreover, the calculated error is relatively high. This is mostly due to the problem is apparatus. Also may be due to the lose connections.